

HW3 HLS Scheduling

Problem 1.

Consider the CDFG below computing:

$$\text{Result} = (x_1 + x_2) - (x_1 \times 2)$$

Operation	Type	Inputs	Delay (cycles)	Dependencies
A	Addition	x_1, x_2	1	Inputs available at $t=0$
B	Multiplication	x_1, const	2	Input x_1 available at $t=0$
C	Subtraction	Out(A), Out(B)	1	Requires A and B outputs

Time model: Operation started at time t with delay d completes at $t + d$; result available for dependent ops at $t + d$.

- Draw the CDFG according to the table above (2 pts).
- ASAP Scheduling** (6 pts)
 - Compute earliest start time for each operation.
 - Tabulate: Operation | Start Time | Completion Time
 - Draw a schedule diagram as in lecture notes (time steps 0–3).
 - Report minimum latency (cycles from start to C completion).
- ALAP Scheduling** (6 pts)
 - Use latency from Task 1 as target.
 - Compute latest valid start time for each operation *without increasing latency*.
 - Tabulate: Operation | Start Time | Completion Time
 - Draw a schedule diagram as in lecture notes (time steps 0–3).
- Mobility Analysis** (6 pts)
 - For each operation: $\text{Mobility} = \text{ALAP_start} - \text{ASAP_start}$

- Identify operations on the critical path. Justify.
- Explain what zero mobility implies for scheduling flexibility.

e) **Critical Thinking** (3 pts)

*If hardware constraints allow only **one arithmetic unit** per cycle:*

- Which schedule (ASAP or ALAP) is immediately feasible?
- Why might neither schedule be optimal under resource constraints?
- What technique would you use next? (Name one algorithm.)